Humberto Sandmann

Computer Scientist & Professor

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São Paulo, Brazil | Open to relocation (on-site)

PROFILE

Computer scientist and professor with a D.Sc. from the University of São Paulo and a research stay at the Max Planck Institute for Dynamics and Self-Organization. My research bridges bio-inspired neural computing, nonlinear dynamics, and pattern recognition, investigating how biological neuronal mechanisms can inform computational intelligence. More than a decade of university teaching alongside senior industry practice in software architecture, artificial intelligence, and high-performance computing.

RESEARCH INTERESTS

- DeFi, smart contracts, and decentralized applications
- Computation in large-scale systems
- Nonlinear systems and network dynamics
- Small-world networks and complex systems
- Artificial intelligence and deep learning
- Synchronization of coupled oscillators

EDUCATION

D.Sc., Electrical Engineering 2007–2012
Polytechnic School, University of São Paulo
Thesis Spike patterns and computation in dynamical neural networks
Advisor Prof. Dr. Emílio Del Moral Hernandez
DOI 10.11606/T.3.2012.tde-05092012-165022

D.Sc. Research Internship 2009–2010
Max Planck Institute for Dynamics and Self-Organization · Göttingen, Germany
Advisor Prof. Dr. Marc Timme
Note Networks Dynamics Group — CAPES grant BEX 4076-09-5

M.Sc., Electrical Engineering 2003–2006
Polytechnic School, University of São Paulo
Thesis Prediction of time series using architecture based on neuro-fuzzy systems
Advisor Prof. Dr. Marco Túlio Carvalho de Andrade
DOI 10.11606/D.3.2006.tde-01042009-095125

B.Sc., Computer Science 1999–2002
Faculdade de Engenharia Industrial (FEI) · São Bernardo do Campo, Brazil

Technician, Industrial Computation 1996–1998
Escola Técnica Jorge Street · São Caetano do Sul, Brazil

ACADEMIC APPOINTMENTS & TEACHING

Insper *Associate Professor (2020–present), Assistant Professor (2018–2020)* 2018–present
Platform, Microservices and APIs · Machine Learning · Neural Networks and Deep Learning · Cloud Computing · Design of Software · Coding Interviews · Foundations for Physical Communication

ESPM *Associate Professor II* 2014–2026
Machine Learning · Applied Mathematics · Object-Oriented Programming · Software Quality · Statistics · Architecture and Operating Systems · Artificial Intelligence and Society

Link School of Business *Associate Professor (freelance)* 2024

IBTA — Instituto Brasileiro de Tecnologia Avançada *Professor* 2016–2017
Basics of Robotics · Object-Oriented Programming · Data Structures and Algorithms

FIAP *Associate Professor* 2012–2015
Distributed Systems · MBA: Big Data & NoSQL Modelling · Data Structures and Algorithms · Databases (Oracle PL/SQL)

Toledo Prudente Centro Universitário *Visiting Professor* 2015
MBA: Big Data and Machine Learning

Universidade Cruzeiro do Sul *Associate Professor* 2012–2013
Computer Architecture · Software Engineering · Mobile Programming (Android) · Databases

Universidade Nove de Julho *Temporary Professor* 2011
Computer Architecture · Introduction to Programming · Web Programming

Polytechnic School, University of São Paulo *Teaching Assistant* 2008–2011
Experimental Pattern Recognition, Modeling and Neurocomputing · Modeling for Signal Processing: Neural Networks and Learning · Foundations of Computing Engineering Laboratory

PUBLICATIONS

[5] **H. Sandmann**, L. A. Silva, E. Del-Moral-Hernandez. A self-organizing architecture of recursive elements for continuous learning (poster abstract). *Dynamics Days Europe, Göttingen, Germany*, 2009, pp. 246–248.

[4] E. Del-Moral-Hernandez, **H. Sandmann**, G. Araujo. Context dependent pattern recognition — A framework for hybrid architectures bridging chaotic neural networks based on Recursive Processing Elements and symbolic information. *Proc. International Joint Conference on Neural Networks (IJCNN), Atlanta, GA, USA*, 2009, pp. 663–670. doi: 10.1109/IJCNN.2009.5179061

[3] L. A. da Silva, **H. Sandmann**, E. Del-Moral-Hernandez. A self-organizing architecture of recursive elements for continuous learning. *Proc. IJCNN (IEEE World Congress on Computational Intelligence), Hong Kong, China*, 2008. doi: 10.1109/IJCNN.2008.4634190

[2] **H. Sandmann**, M. T. C. de Andrade. Análise da capacidade de uma rede neural de memória associativa com arquitetura de elementos processados recursivamente. *Proc. Brazilian Symposium on Neural Networks (SBRN), São Luís, MA, Brazil*, 2004.

[1] E. Del-Moral-Hernandez, **H. Sandmann**, L. A. da Silva. Pattern recovery in networks of recursive processing elements with continuous learning. *Proc. IJCNN, Budapest, Hungary*, 2004. doi: 10.1109/IJCNN.2004.1379877

GRANTS & FUNDING

CNPq	Doctoral fellowship — grant 140632/2007-7	2007–2011
CAPES	Doctoral internship abroad (Max Planck Institute) — BEX 4076-09-5	2010
CAPES (PAE)	Teaching-assistantship fellowship — Polytechnic School, USP	2008, 2011
FAPESP	Teaching-assistantship fellowship — Polytechnic School, USP	2008

CONFERENCE SERVICE

IJCNN 2020	Reviewer · Glasgow, Scotland
IJCNN 2019	Program Committee & Reviewer · Budapest, Hungary
IJCNN 2018	Reviewer · Rio de Janeiro, Brazil
IJCNN 2017	Program Committee & Reviewer · Anchorage, AK, USA
IJCNN 2015	Reviewer · Killarney, Ireland

INDUSTRY EXPERIENCE

H-Sandmann	Founder & Consulting IT Specialist	2006–2025
SelsanTech	Co-Founder & Research Scientist	2012–2023
RockSpoon	Senior Software Architect & Machine Learning Engineer	2018–2019
Banco Votorantim	Senior Consultant, Computational Optimization	2015–2016
TecBan	Senior Consultant, Computational Optimization	2011
ABN Amro Bank	Senior Software Engineer / Architect	2006–2007
Politec	Senior System Analyst	2005–2006
EverSystems	Senior Software Engineer / Architect	2004–2005
Central Online	IT Project Manager	2002–2003
Datamace	Senior System Analyst	1999–2002
Heraeus Electro-Nite	Computer Programmer	1998–1999

PROFESSIONAL MEMBERSHIPS

IEEE — Institute of Electrical and Electronics Engineers Member #60216475

- Computational Intelligence Society (CIS) — member since 2004; Brazilian committee (2009; 2010–2012)
- Signal Processing Society (SPS) — 2005–2009
- Engineering in Medicine and Biology Society (EMBS) — 2005–2009

INNS — International Neural Network Society Student member #3951

LANGUAGES

Portuguese Native **English** Advanced **German** B1